

培道中學校

姓名：

(號)

成績

年級：

科目：

練習題 16.8

2.

$$\int_0^1 dx_1 \int_0^{x_1} dx_2 \cdots \int_0^{x_{n-1}} x_1 x_2 \cdots x_{n-1} x_n dx_n = \frac{1}{2} \int_0^1 dx_1 \int_0^{x_1} dx_2 \cdots \int_0^{x_{n-2}} x_1 x_2 \cdots x_{n-2} x_{n-1}^3 dx_{n-1}$$

$$= \frac{1}{2 \times 4} \int_0^1 dx_1 \int_0^{x_1} dx_2 \cdots \int_0^{x_{n-3}} x_1 x_2 \cdots x_{n-3} x_{n-2}^5 dx_{n-2} = \cdots = \frac{1}{(2n-2)!} \int_0^1 x_1^{2n-1} dx_1 = \frac{1}{(2n)!}$$

$$3. \int_{V_n} dx_1 \cdots dx_n = \int_{V_n} a_1 a_2 \cdots a_n d\left(\frac{x_1}{a_1}\right) d\left(\frac{x_2}{a_2}\right) \cdots d\left(\frac{x_n}{a_n}\right) = a_1 a_2 \cdots a_n S_n(1) = \frac{1}{n!} a_1 a_2 \cdots a_n$$

$$b) \int_{V_n(a)} dx_1 dx_2 \cdots dx_n = 2^n S_n(a) = \frac{2^n}{n!} a^n = \frac{(2a)^n}{n!}$$

$$4. K_{m+n+1}(x, y) = \int_{[a,b]^{m+1}} K(x, t_1) K(t_1, t_2) \cdots K(t_m, t_{m+1}) K(t_{m+1}, t_{m+2}) \cdots K(t_{m+n+1}, y) dt_1 dt_2 \cdots dt_{m+n+1}$$

$$= \int_{[a,b]^m} K(x, t_1) K(t_1, t_2) \cdots K(t_m, t_{m+1}) dt_1 dt_2 \cdots dt_m \int_{[a,b]} K(t_{m+1}, t_{m+2}) K(t_{m+2}, t_{m+3}) \cdots K(t_{m+n+1}, y) dt_{m+2} dt_{m+3} \cdots dt_{m+n+1}$$

$$= \int_a^b K_m(x, t_{m+1}) K_n(t_{m+1}, y) dt_{m+1} = \int_a^b K_m(x, t) K_n(t, y) dt$$

問題 16.8

$$1. \int_0^a dx_1 \int_0^{x_1} dx_2 \cdots \int_0^{x_{n-1}} f(x_n) dx_n = \int_{0 \leq x_n \leq x_{n-1} \leq x_{n-2} \leq \cdots \leq x_1 \leq a} f(x_n) dx_1 dx_2 \cdots dx_n$$

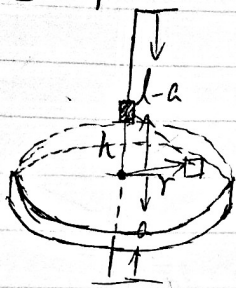
$$= \int_0^a dx_n \int_{x_n}^a dx_{n-1} \int_{x_{n-1}}^a dx_{n-2} \cdots \int_{x_2}^a f(x_n) dx_1 = \int_0^a f(x_n) \left(\int_{x_n \leq x_{n-1} \leq \cdots \leq x_1 \leq a} dx_{n-1} dx_{n-2} \cdots dx_1 \right) dx_n$$

$$= \int_0^a f(x_n) \left(\int_{x_n \leq x_{n-1} \leq \cdots \leq x_2 \leq a} (a - x_2) dx_{n-1} \cdots dx_2 \right) dx_n = \int_0^a f(x_n) \left(\frac{1}{2!} \int_{x_n \leq x_{n-1} \leq \cdots \leq x_3 \leq a} (a - x_3)^2 dx_{n-1} \cdots dx_3 \right) dx_n$$

$$= \cdots = \frac{1}{(n-1)!} \int_0^a f(x_n) (a - x_n)^{n-1} dx_n = \frac{1}{(n-1)!} \int_0^a f(t) (a - t)^{n-1} dt$$

練習題 16.9

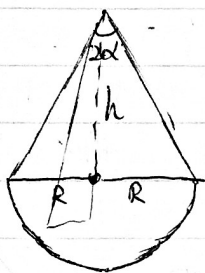
2.



$$\begin{aligned}
 F_z &= \int_{-a}^{l-a} \rho dz \int_{x^2+y^2 \leq R^2} \frac{\mu dx dy}{x^2+y^2+z^2} \cdot \frac{-z}{\sqrt{x^2+y^2+z^2}} \\
 &= \int_{-a}^{l-a} \rho dz \int_0^{2\pi} d\theta \int_0^R \frac{-\mu z r}{(r^2+z^2)^{\frac{3}{2}}} dr \\
 &= 2\pi \int_{-a}^{l-a} \rho \frac{\mu z}{\sqrt{r^2+z^2}} \Big|_0^R dz = 2\pi \int_{-a}^{l-a} \left(\frac{\mu z}{\sqrt{R^2+z^2}} - \mu \right) dz \\
 &= 2\pi \mu \sqrt{R^2+z^2} \Big|_{-a}^{l-a} - 2\pi \mu l = 2\pi \mu \left(\sqrt{R^2+(l-a)^2} - \sqrt{R^2+a^2} - l \right)
 \end{aligned}$$

由對稱性知 $F_x = F_y = 0$, 所以 $F = F_z = 2\pi \mu \left(\sqrt{R^2+(l-a)^2} - \sqrt{R^2+a^2} - l \right)$

3.



$$\begin{aligned}
 F_z &= \int_{\substack{z \leq 0 \\ x^2+y^2+z^2 \leq R^2}} \frac{\rho d\mu}{x^2+y^2+(h-z)^2} \cdot \frac{z-h}{\sqrt{x^2+y^2+(h-z)^2}} + \int_{\substack{z > 0 \\ x^2+y^2 \leq (h-z)^2 \tan^2 \alpha}} \frac{\rho d\mu}{x^2+y^2+(h-z)^2} \cdot \frac{z-h}{\sqrt{x^2+y^2+(h-z)^2}} \\
 &= \int_0^{2\pi} d\varphi \int_{\frac{\pi}{2}}^{\pi} d\theta \int_0^R \frac{\rho (r \cos \theta - h) r^2 \sin \theta}{(r^2+h^2-2rh \cos \theta)^{\frac{3}{2}}} dr + \int_0^h dz \int_0^{2\pi} d\theta \int_0^{(h-z) \tan \alpha} \frac{\rho (z-h)}{(r^2+(h-z)^2)^{\frac{3}{2}}} r dr \\
 &= 2\pi \int_0^R dr \int_{\frac{\pi}{2}}^{\pi} \frac{\rho (h-r \cos \theta) r^2}{(r^2+h^2-2rh \cos \theta)^{\frac{3}{2}}} d(\cos \theta) + 2\pi \int_0^h dz \int_0^{(h-z) \tan \alpha} \frac{\rho (z-h) r}{(r^2+(h-z)^2)^{\frac{3}{2}}} dr
 \end{aligned}$$

記 $G(r) = \int_0^{\pi} \frac{h-rx}{(r^2+h^2-2rhx)^{\frac{3}{2}}} dx$

設 $t^2 = r^2+h^2-2rhx \Rightarrow 2tdt = -2rh dx \Leftrightarrow dx = -\frac{1}{rh} t dt$

$$G(r) = \int_{\sqrt{r^2+h^2}}^{\sqrt{r^2+h^2}} \frac{h - \frac{t^2-r^2-h^2}{-2h}}{t^{\frac{3}{2}}} \cdot \left(-\frac{1}{rh}\right) t dt = -\frac{1}{2rh^2} \int_{\sqrt{r^2+h^2}}^{r+h} \frac{2h^2+t^2-r^2-h^2}{t^2} dt$$

$$= -\frac{1}{2rh^2} \int_{\sqrt{r^2+h^2}}^{r+h} \left(\frac{h^2-r^2}{t^2} + 1 \right) dt = -\frac{1}{2rh^2} \left(\frac{r^2-h^2}{t} + t \right) \Big|_{\sqrt{r^2+h^2}}^{r+h}$$

$$= -\frac{1}{2rh^2} \left(2r - \frac{r^2}{\sqrt{r^2+h^2}} \right) = -\frac{1}{h^2} \left(1 - \frac{r}{\sqrt{r^2+h^2}} \right) = \frac{1}{h^2} \left(\frac{r}{\sqrt{r^2+h^2}} - 1 \right)$$

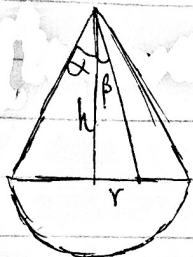
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$$r = h \tan \beta, \quad G(r) = \frac{1}{h^2} (\sin \beta - 1)$$

$$\int_0^R \rho r^2 G(r) dr = \int_0^\alpha \rho \cdot h^2 \tan^2 \beta \cdot \frac{1}{h^2} (\sin \beta - 1) \cdot h (\tan^2 \beta + 1) d\beta$$

$$= \rho h \int_0^\alpha \tan^2 \beta (\sin \beta - 1) d \tan \beta = \frac{\rho h}{3} \left[(\sin \beta - 1) \tan^3 \beta \Big|_0^\alpha - \int_0^\alpha \tan^3 \beta \cos \beta d\beta \right]$$

$$= \frac{\rho h}{3} \left[(\sin \alpha - 1) \tan^3 \alpha - \int_0^\alpha \tan^2 \beta \sin \beta d\beta \right]$$

$$= \frac{\rho h}{3} \left[(\sin \alpha - 1) \tan^3 \alpha + \int_0^\alpha \left(\frac{1}{\cos^2 \beta} - 1 \right) d(\cos \beta) \right]$$

$$= \frac{\rho h}{3} \left[(\sin \alpha - 1) \tan^3 \alpha + \left(-\frac{1}{\cos \beta} + \cos \beta \right) \Big|_0^\alpha \right]$$

$$= \frac{\rho h}{3} \left[(\sin \alpha - 1) \tan^3 \alpha - \left(\frac{1}{\cos \alpha} + \cos \alpha - 2 \right) \right]$$

$$= \frac{\rho R}{3} \left[(\sin \alpha - 1) \tan^2 \alpha - \frac{1}{\sin \alpha} - \frac{\cos \alpha}{\tan \alpha} + \frac{2}{\tan \alpha} \right]$$

$$\int_0^h dz \int_0^{(h-z)\tan \alpha} \frac{\rho(z-h)r}{(r^2+(h-z)^2)^{\frac{3}{2}}} dr = \int_0^h \frac{\rho(h-z)}{\sqrt{r^2+(h-z)^2}} \Big|_0^{(h-z)\tan \alpha} dz$$

$$= \int_0^h \rho (\cos \alpha - 1) dz = \rho h (\cos \alpha - 1) = \frac{\rho R}{\tan \alpha} (\cos \alpha - 1)$$

$$\begin{cases} F_z = \frac{2\pi \rho R}{3} \left(\sin \alpha \tan^2 \alpha - \tan^2 \alpha - \frac{1}{\sin \alpha} + \frac{2 \cos \alpha}{\tan \alpha} - \frac{1}{\tan \alpha} \right) = F \\ F_x = F_y = 0 \end{cases}$$

$$4. M_x = \frac{1}{M} \int_{\frac{a}{2} \leq x^2+y^2 \leq a^2} x \rho(x,y) dx dy = \frac{1}{M} \int_0^{2\pi} d\theta \int_{\frac{a}{2}}^a r \cos \theta \cdot r \cdot r dr = \frac{1}{M} \left(\int_0^{2\pi} \cos \theta d\theta \right) \left(\int_{\frac{a}{2}}^a r^3 dr \right) = 0$$

同理可得 $M_y = 0$.

所以重心在樁是圓盤圓心處。

練習題 17.1

$$1. (3) ds = \sqrt{x'^2 + y'^2 + z'^2} dt = \sqrt{(\cos t - t \sin t)^2 + (\sin t + t \cos t)^2 + 1^2} dt$$

$$= \sqrt{1^2 + 1^2 + t^2} dt = \sqrt{t^2 + 2} dt$$

$$\int_I z ds = \int_0^{2\pi} t \sqrt{t^2 + 2} dt = \frac{1}{2} \int_0^{2\pi} \sqrt{t^2 + 2} dt^2 = \frac{1}{3} (t^2 + 2)^{\frac{3}{2}} \Big|_0^{2\pi} = \frac{1}{3} [(4\pi^2 + 2)^{\frac{3}{2}} - 2^{\frac{3}{2}}]$$

$$(5) ds = \sqrt{a^2(1 - \cos t)^2 + a^2 \sin^2 t} dt = a \sqrt{2 - 2 \cos t} dt = \sqrt{2} a \sqrt{1 - \cos t} dt$$

$$\int_I y^2 ds = \int_0^{2\pi} a^2(1 - \cos t)^2 \cdot \sqrt{2} a \sqrt{1 - \cos t} dt = \sqrt{2} a^3 \int_0^{2\pi} (2 \sin^2 \frac{t}{2})^2 \cdot \sqrt{2} \sin \frac{t}{2} dt$$

$$= 8a^3 \int_0^{2\pi} \sin^5 \frac{t}{2} dt = 16a^3 \int_0^{\pi} \sin^5 t dt = -16a^3 \int_0^{\pi} (1 - \cos^2 t)^2 d(\cos t)$$

$$= -16a^3 \int_1^{-1} (1 - 2x^2 + x^4) dx = 16a^3 \int_{-1}^1 (x^4 - 2x^2 + 1) dx = 32a^3 \left(\frac{1}{5} - \frac{2}{3} + 1 \right) = \frac{256}{15} a^3$$

$$2. \text{ 記 } x = a \cos t, y = b \sin t, t \in [0, 2\pi]$$

$$\text{周長} = \int_0^{2\pi} \sqrt{a^2 \sin^2 t + b^2 \cos^2 t} dt = a \int_0^{2\pi} \sqrt{1 - (1 - \frac{b^2}{a^2}) \cos^2 t} dt = a \int_0^{2\pi} \sqrt{1 - e^2 \cos^2 t} dt$$

$$3. x = f(\theta) \cos \theta, y = f(\theta) \sin \theta$$

$$\int_I ds = \int_{\alpha}^{\beta} \sqrt{x'(\theta)^2 + y'(\theta)^2} d\theta = \int_{\alpha}^{\beta} \sqrt{(f'(\theta) \cos \theta - f(\theta) \sin \theta)^2 + (f'(\theta) \sin \theta + f(\theta) \cos \theta)^2} d\theta$$

$$= \int_{\alpha}^{\beta} \sqrt{f'(\theta)^2 + f(\theta)^2} d\theta$$

$$\text{橢圓: } r^2 = \frac{a^2 b^2}{a^2 \sin^2 \theta + b^2 \cos^2 \theta} \Leftrightarrow r = \frac{ab}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}} = \frac{b}{\sqrt{1 - e^2 \cos^2 \theta}}, r'(\theta) = \frac{-b e^2 \cos \theta \sin \theta}{(1 - e^2 \cos^2 \theta)^{\frac{3}{2}}}$$

$$\int_I ds = \int_0^{2\pi} \frac{b}{\sqrt{1 - e^2 \cos^2 \theta}} \sqrt{1 + \frac{e^4 \cos^2 \theta \sin^2 \theta}{(1 - e^2 \cos^2 \theta)^2}} d\theta$$